MIL-P-46121B(MR)
12 September 1977
SUPERSEDING
MIL-P-46121A(MR)
18 November 1971

MILITARY SPECIFICATION

PLASTIC SHEET AND COATING MATERIAL, PARA-XYLYLENE POLYMERS

This specification is approved for use by the Army Materials and Mechanics Research Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers para-xylylene polymeric material as thin sheet or as a continuous coating.

1.2 Classification. The polymers shall be of the following types and classes, as specified (see 6.2).

Type I - High melting point
Type III - Low melting point
Class 1 - Sheets
Class 2 - Coating materials

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL
PPP-B-585 - Boxes, Wood Wirebound
PPP-B-601 - Boxes, Wood, Cleated-Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock Corner
PPP-B-636 - Box, Fiberboard

STANDARDS
MIL-P-116 - Preservation, Methods of

FSC 9330

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Army Materials and Mechanics Research Center, Watertown, MA 02172 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 150 - AC Loss Characteristics and Dielectric constant (Permittivity) of Solid Electrical Insulating Materials
D 618 - Conditioning Plastics and Electrical Insulating Materials for Testing
D 792 - Specific Gravity and Density of Plastics by Displacement
D 882 - Tensile Properties of Thin Plastic Sheeting
D 883 - Definitions of Terms Relating to Plastics
E 96 - Water Vapor Transmission of Materials in Sheet Form

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

National Motor Freight Traffic Association, Inc., Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations Inc., Traffic Department, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606).

3. REQUIREMENTS

3.1 Material. The thermoplastic material shall consist of poly-para-xylylene for type I, and poly-monochloro-para-xylylene for type III polymer. These materials are linear, highly crystaline polymers. Reclaimed material is permitted when it meets all requirements specified herein and its molecular weight is essentially the same as that of the unused polymer for the type and class of material specified.

3.2 Property values. The sheet or coated material shall conform to the property values specified in table I, when tested as specified in the applicable procedure of 4.3.

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Table I. Property values[1/]

Property	Type I	Type III
Tensile strength, MPa (psi), min.	41.4 (6000)	62.0 (9000)
Elongation, percent, min.	10	100
Specific gravity	1.1 ± 0.1	1.3 ± 0.1
Melting point, range		
°C	393-404	279-299
°F	740-760	535-570
Water vapor transmission rate, grams per 0.0254 mm (0.001 inch) thickness, per 0.0645 square meter (100 square inch) surface area, per 24 hours, max.	1.7	0.5-0.7
Dissipation factor at 1 kilohertz, max.	0.0003	0.03

[1/] Applicable to class 1 and class 2 material. Metric unit abbreviations: MPa = megapascal, mm = millimetre.

3.3 Color. The color of the sheet or coated material shall be natural.

3.4 Dimensions and tolerances. The length, width thickness, and tolerances for each dimension shall be as specified by the procuring activity.

3.5 Workmanship. The sheet or coated material shall be uniform and essentially free from cracks, scratches, bubbles, imbedded particles and other defects that affect appearance or which may affect serviceability. (These defects may be defined in accordance with ASTM D 883, as applicable.)

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.2 Sampling for inspection and acceptance. Sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type submitted for inspection at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of this specification and, to the extent specified, of all referenced subsidiary specifications and standards. In the event of conflict, this specification shall govern. A supplier's certificate of compliamce with 3.1 shall be furnished.

4.2.2 Inspection of the sheet or coated material. Examination shall be made in accordance with the classification of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purposes of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of packages of plastic sheets or coated shapes as applicable, for examination in 4.2.2.1, 4.2.2.2, and in units of shipping containers for examination in 4.2.2.3.

4.2.2.1 Examination of the sheet or coated material for defects in appearance, construction, and workmanship. The sample unit for this examination, specified in table II, shall be one sheet or one coated shape, as applicable. Not more than five sample units shall be taken from any one package of sheets or coated shapes.

Table II. Examination for defects in appearance, construction and workmanship

Examine	Defect
Appearance	Not clean, presence of imbedded particles, dirt, grit or other foreign matter, that may affect appearance or serviceability.
	Material not natural in color.
Construction and workmanship	Any cracks, scratches, bubbles, warpage, pits or other defects that would affect serviceability.
	Any cut, puncture, sharp crease, wrinkle, tear or hole.
	Edges not clean cut; ragged, crushed or uneven edges.

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4.2.2.2 Examination of the sheet or coated material for dimensional defects. The sample unit for this examination specified in table III shall be one plastic sheet, or one coated shape, as applicable.

Table III. Examination for dimensional defects

Examine	Defect
Length and width	Not as specified. Varies by more than length and width tolerances specified by the procuring activity.
Thickness	Not as specified. Varies by more than the tolerance specified by the procuring activity.

4.2.2.3 Examination of packaging. An examination in accordance with table IV, shall be made to determine that preservation, packing, and marking comply with section 5 requirements. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Table IV. Examination of packaging

Examine	Defect
Preservation	Not level specified. Not unit packed, multiple packed, or intermediate packed, as specified. Unit packing material not as specified: Closure not accomplished by specified or required methods or materials; component missing. Weight of intermediate packing exceeds specified requirements.
Packing	Not level specified. Containers not as specified, closures not accomplished by specified or required methods or materials. Any nonconforming component, component missing, damaged or otherwise defective, affecting serviceability.
Count	Less than specified or indicated quantity.
Weight	Gross weight exceeds specified requirements.
Markings	Interior or exterior markings, as applicable, omitted, illegible, incorrect, or improper size, location or sequence or method of application. Not in accordance with contract requirements.

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4.2.2.4 Inspection levels and AQLs for examination. The inspection levels for determining the sample size and the AQL, expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1	I	2.5
4.2.2.2	S-2	4.0
4.2.2.3	S-2	2.5

4.2.3 Classification of tests. All tests under this specification shall be classified as lot acceptance tests. They shall be made on each lot of material offered against this specification and, in conjunction with the above examination, shall be the basis of acceptance or rejection of the lot.

4.2.4 Testing. The sheet or coated material shall be tested for the applicable characteristics listed in table I in accordance with the test method specified herein for each lot submitted for inspection. Sampling shall be in accordance with MIL-STD-105, and the lot size for the purpose of determining sample size for testing shall be expressed in units of packages of sheet or coated material, as applicable. The inspection level shall be S-1. The sample size (number of units of product drawn from a lot) shall be in accordance with the single sampling plan for normal inspection and each unit shall consist of sufficient material to prepare all specimens required for testing. Each sample unit shall be tested and must meet all applicable requirements of section 3. Test results for each characteristic in each sample unit shall be the average results from the specimens tested for that characteristic.

4.3 Test methods.

4.3.1 Test specimens. Test specimens shall be prepared in accordance with the recommendation of the manufacturer.

4.3.2 Conditioning and testing. Test specimens shall be conditioned in accordance with procedure A of ASTM D 618 at a temperature of 23° ± 2°C (73.4° ± 3.6°F). Testing shall be at 23° ± 2°C (73.4° ± 3°F) and 50 ± 5 percent relative humidity except for the specific gravity determination which shall be in accord with the test method specified (see 4.3.4).

4.3.3 Tensile strength and elongation. Three specimens shall be tested in accordance with method A of ASTM D 882. The speed of testing shall be between 0.508 and 0.635 mm (0.20 and 0.25 inches) per minute.

4.3.4 Specific gravity. One specimen shall be tested in accordance with ASTM D 792, using method A-1.

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4.3.5 Melting point. One specimen shall be tested using a Fisher-Johns, or equal, melting point apparatus with appropriate thermometer and cover glass. The equipment shall be operated at 110 volts and 50 to 60 hertz.

4.3.6 Water vapor transmission rate. Three specimens shall be tested in accordance with procedure E of ASTM E 96, except that the calculation shall be based on grams per 0.0254 mm (0.001 inch) thickness, per (0.0645 square meter (100 square inch) surface area, per 24 hours.

4.3.7 Dissipation factor. Three specimens shall be tested in accordance with ASTM D 150 at a frequency of one kilohertz.

5. PACKAGING

Application. The requirements of section 5 apply only to purchase by or direct shipment to the Government.

5.1 Preservation shall be level A, B, or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packing. Unless otherwise specified in the contract or order (see 6.2), sheets and coated shapes shall be unit packed in quantities specified by the procuring agency in accordance with method III of MIL-P-116. Shapes of only one set of nominal dimensions shall be placed in one package. Individual sheets shall be masked or unmasked at the option of the procuring agency.

5.1.1.2 Intermediate packing. When required, specified quantities of unit packs shall be intermediately packed as specified in the contract or purchase order (see 6.2).

5.1.2 Level B. Preservation shall be the same as for level A (see 5.1.1).

5.1.3 Level C. Sheets and coated shapes shall be preserved in accordance with commercial practice to provide adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. Sheets and coated shapes prepared as specified in 5.1.1 shall be separated by thickness, and shall be packed in containers conforming to any of the following specifications, at the option of the supplier:

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Specification	Type or class
PPP-B-585	Class 3 use
PPP-B-601	Overseas type
PPP-B-621	Class 2

Boxes shall be closed, strapped, or banded in accordance with the applicable box specification or appendix thereto. The gross weight of the shipping container shall be as specified in the applicable container specification.

5.2.2 Level B. Sheets and coated shapes shall be packed in domestic class or type shipping containers conforming to PPP-B 585, PPP-B-601, PPP-B-621, or PPP-B-636 class weather resistant. All containers shall be closed in accordance with the applicable container specification or appendix thereto. The gross weight of the shipping container shall be as specified in the applicable container specification.

5.2.3 Level C. Packing shall be in accordance with commercial practice adequate to ensure acceptance and delivery by the carrier for the mode of transportation employed. Containers shall comply with the requirements of the Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable to the mode of transportation.

5.3 Marking. In addition to any special marking specified in the contract or order, containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. Para-xylylene polymers are intended for use where properties required include outstanding dielectrics and electrical insulation, insolubility in organic substances up to 149°C (300°F), good gas and moisture barriers, and good dimensional stability to changes in relative humidity. Cycling from approximately -271°C (-455°F) to room temperature does not affect electrical properties. Intended uses include coatings that provide dielectric and electrical insulation, chemical protection of reactive materials, and protection from the environment. The sheeting is intended, also, for optical applications.

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6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and class of material (see 1.2).
c. Sheet length, width, thickness, and tolerances required (see 3.4).
d. Selection of applicable levels of preservation and packing required (see 5.1 and 5.2).
e. Quantities required in unit packing (see 5.1.1.1).
f. Quantities for intermediate packing, if required (see 5.1.1.2).
g. Special marking, if required (see 5.3).

6.3 Supersession data. Type II material of MIL-P-46121(MR) has been deleted.

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MIL-P-46121B(MR), Plastic Sheet and Coating Material, Para-Xylylene Polymers

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